(Lecture : 1)

CSE 422

AI ? Artificial Inteligence (term invented in 1950 get recognized in 1956)

↓

→ Remain in a Agent → Complex cognitive ability
→ Use of Experience
→ Self awerness
→ Decision Making

Agent
- → input
- → process
- → output

AI : (Pillars)

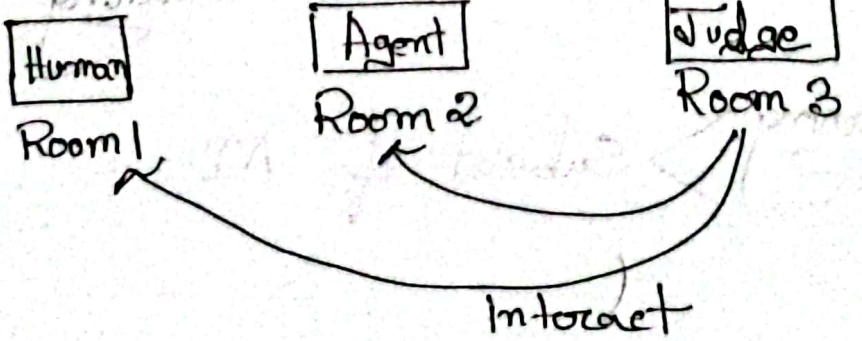
→ Store knowledge (Experience)

→ Reason about knowledge (Accomplish a task using knowledge)

→ Behave intelligently (Any complex decision making)

→ Interact with people, agents and environment

Turing Test :



Judge interact with human and agent

→ if agent can give or can detect who's agent correctly → agent is not smart

→ if judge is incorrect / confused → agent is smart and intelligent

1970 (First AI winter) : → Not effective AI

1980 (2nd AI winter) : "

After 2000 progress of AI start

CNN — Convolutional Neural Network

→ Alexnet (Deep learning framework) (use for image classification & achive high accuracy)

→ VGG

→ Inception

AI is not same as Machine Learning

Machine learning > Subset of AI

Deep " >

Deep Learning → has no manual implementation

Dynamical learning

Due to AI Progess

→ lost of job

→ AI rationality (Try to achive goal in most efficient way but it would be wrong / unethical)

→ Stop button

- (may not work)

Behavior:

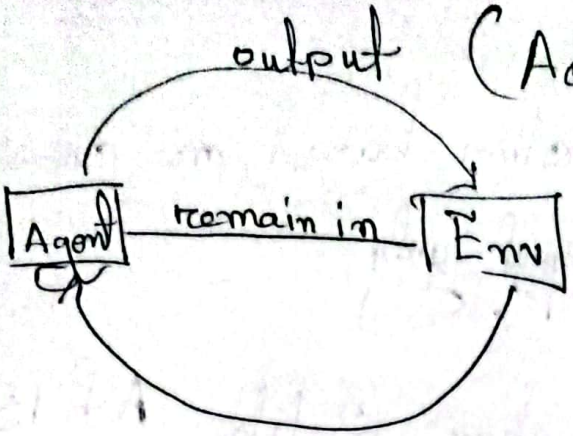
→ Human like behavior (think and use emotion)

→ Rational " (just do thinks to achieve higest profit. will not consider any other ethical or emotional issue)

Lecture - 2

How to design an intelligent agent?

- An intellagent agent -
  - percieve environment via sensors
  - act rationally with its actuators
- A discrete agent -
  - recieve percepts one at a time
  - maps this percept sequence to a sequence of discrete actions.
- Properties
  - Autonomus.
  - Reactive to the environment
  - Pro-active
  - Interact with other agents



(Lecture - 2)

Input can be → eyes, ears etc

Input take through module → sensors

Output generate " " → Actuators

Same element may work as Actuator & Sensors

positive impact → performance high else low

(Sense, Actuator, Env, Performance measurement)

(PEAS)

Agent < simple (light on off on basis of day light)
complex (Self driving car) → module complex
pipeline "
processing fuction "

i) **Simple reflex agent** → Action based on current input only.

(table driven agent)

- Vacume cleaner to clear 2-tile A & B

⇒ Use action table to find the next action

Perception	Action
A dusty	A clean
A clean	Go B
B dusty	Clean B
B clean	Go A

(1) ⇒ Cannot deal with continuously changing environment (No scalable)

(2) ⇒ Agent is not dynamic for every change in environment need to change the agent

⇒ Agent here is static

(3) ⇒ So at complex env it will not work

⇒ For complex env, perception action table can go to infity no of action

(4) ⇒ Prone to looping

Disadvantage

⇒ Not scalable
⇒ Not suitable for long env
⇒ loop

ii) Agent with memory / Model based agent:

- A clean → set to memory its clean
- Memory is dynamic
- Have internal state that is used to keep track of past state.

iii) Goal Based Agent: (Rational behavior):

Act on the basis of goal rather than performing action on the ~~goal~~ based perception table - have goal information that describe desirable situations.

Input → Configuration of perception and action

goal → Number of dusty tile 0

no of dusty tile = 2

heuristic

↓

A particular agent is at what distant from its targeted goal

iv) Utility Based Agent :

Multimodal heuristic

Utility fuction → side goal (may be for safety)

To act goal → has to achieve (Primary)

utility → not necessary to achieve

Happiness Score → from ulity function

Peformance evalution matrix → low.

- take deision on classic axiomatic utility theory.

Properties of Environment :

i) Partially observable vs fully observable

(fully observe করে কিনা sensor)

ii) Deterministic vs. stochastic:

Deterministic → for one action the output is always correctly predicted / known.

Stochastic → environment not correctly predicted on the basis of action (cannot be)

iii) Episodic vs sequential:

- not linked (action is not linked with others)
- linked

iv) Static vs dynamic:

- no change in environment
- change in environment (Real world ")

Single agent vs Multi agent:

- opponent নাই
- involvement of opponent

vi) Discreet vs continous:

- environment defined using finite no.
- cannot defined using finite number

vii) Known vs Unknown:

- after learning from data

(Lecture-3)

Informed Search (More Efficient)

- Greedy Best First Search (
- A* Search.

8 Puzzle game

Goal State equal -

1	2	3
4	5	6
7	8	

Here we will use heuristic value - ie is how much far it is from goal state

Let Consider -

h = 5

1	2	3
5	↑ ← → ↓	8
4	6	7

→ Start State

→ heurist (h) = No of misplaced tile

h=6

1		3
5	2	8
4	6	7

h=5

1	2	3
5	6	8
4		7

h=4

1	2	3
	5	8
4	6	7

h=4

1	2	3
5	8	
4	6	7

So we will chose whose h value is less - means closer to goal

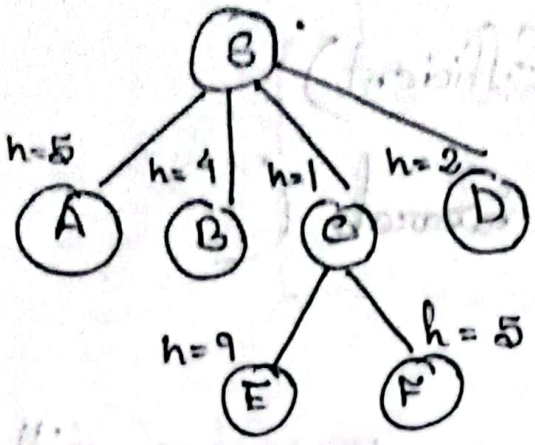
h=5

	2	3
1	5	8
4	6	7

h=3

1	2	3
4	5	8
	6	7

Let a Graph be:



S	5
A	5
B	4
C	1
D	2
E	9
F	5
I	0

→ Goal state

8 puzzle:

- h_1 = No of misplaced tile
- h_2 = Manhattan distance (better)

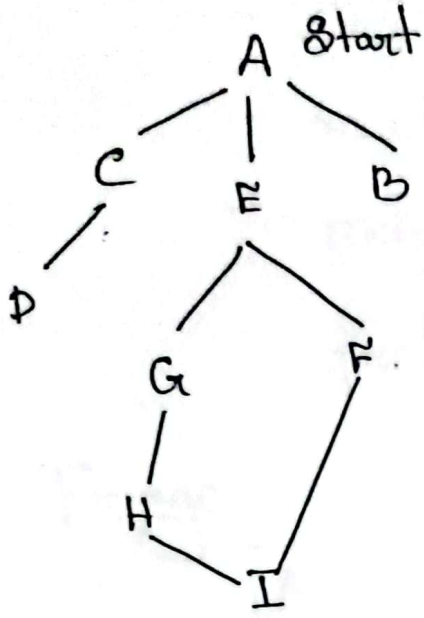
h = at how much distance the misplaced tile is.

1	3	2
4	7	8
6	5	

3 2 7 8 6 5
↓ ↓ ↓ ↓ ↓ ↓
$h = 1 + 1 + 2 + 2 + 3 + 1$
$= 10$

- Manhattan distance value is always large
  ∴ larger the better ∴ So: h_2 is better.
- Here h_2 dominate h_1
- ie if $h_1 \geq h_2$; h_1 dominates h_2

i) Greedy Best First Search :



A	366
B	374
C	329
D	244
E	256
F	178
G	193
H	98
I	0

Fringe : (Priority Queue)

A^{366}

A^{366} | C^{329} E^{253} B^{374}

(here lowest edge value will be pop)

lowest h value ← E^{253} | C^{329} B^{374} G^{193} F^{178}

F^{178} | C^{329} B^{374} G^{193} I^{0}

I^{0} | C^{329} B^{374} G^{193}

(As I^{0} expand & pop So the fringe terminate)

Here path is - $A \rightarrow E \rightarrow F \rightarrow I$

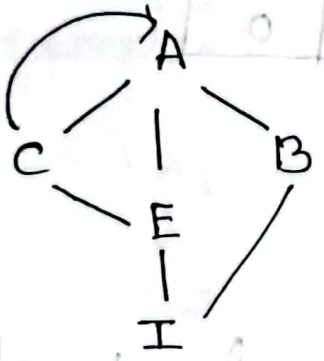
path cost = 450 (Not Optimal) -

As we get less path cost in different way

- GBFS → Not Optimal
  → Complete ?

Search has 2 version
- Tree version
- Giraph version

Tree Version :



A	1
B	3
C	2
E	4
I	0

Fringe :

A¹

A' | C² E⁴ B³

C | A² E⁴ B³

A | E⁴ B³ C² E⁴ - - - -

create loop C and A repeat again and again

- So tree GBFS is not complete (create loop)

Graph Search:

- Used to prevent loop
- mark visited node
- skip the visted node while expanding/ pushing into the fringe.

Fringe

	A'
A'	e^2 E^4 B^3
C^2	E^4 B^3
B^3	I^0
I^0	(terminate)

Q. Complete ?

- Tree version → Incomplete
- Graph version → Finite → complete
  Infinite → incomplete

Time & Space Complexity - $O(b^m)$

b - branching factor
m - max depth

(Lecture - 4)

Informed Search (Single Player Game)

(A* Search)

$$f(n) = g(n) + h(n)$$

$h(n)$ = heuristic value

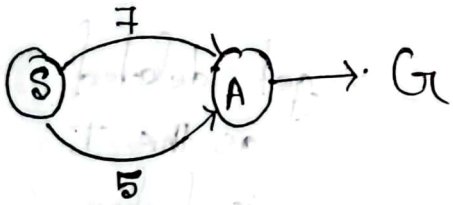
$g(n)$ = Actual path cost from start node (lowest)

For node $n \to F$ (From slide)

$$g(n) + h(n)$$

$$(140 + 99) \quad 178$$

$\therefore f(n) = 417$



$$f(A) = g(A) + h(A)$$

$$\downarrow 5 + (\quad)$$

$f(D) = g(D) + h(D) \to$ (from slide)

$= 229 + 244$

$= 473$

Graph version of A*

Fringe:

A $^{h(A)+g(A) = 366+0 = 366}$

A^{366} B $^{374+78=449}$ C $^{329+118=447}$ E $^{253+140=393}$

E^{393} B^{449} C^{447} G $^{193+(140+80)=413}$ F $^{178+(239)=417}$

G^{413} B^{449} C^{447} F^{417} H $^{98+(140+80+97)=415}$

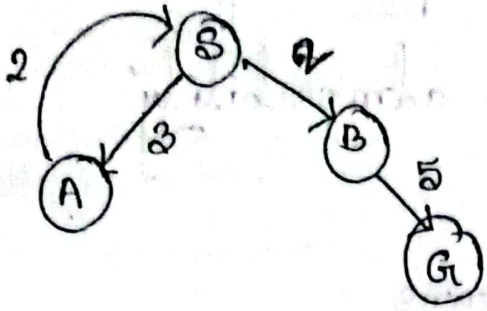
H^{415} B^{449} C^{447} F^{417} I $^{0+418=418}$

F^{417} B^{449} C^{447} I^{418} I $^{0+450=450}$ → get deleted as this I is higher

I^{418} B^{449} C^{447}

I ← H ← G ← E ← A

(∴ A* search is Complete)



S	2
B	7
A	3
G	0

GBFS:

Fringe:

S^2

S^2 | A^3 B^7

A^3 | B^7 S^2

S^2 | B^7 A^3

A^3 | B^7 S^2

fall in loop, so will never reach goal

A* Search:

Fringe:

S^{2+0}

S^2 | $A^{3+3=6}$ $B^{7+2=9}$

A^6 | B^9 $S^{2+(3+2)=7}$

S^7 | B^9 $A^{8+8=11}$

$S^3 \rightarrow A^2 \rightarrow S^2 \rightarrow A$

$\therefore$ (A* search is optimal)

B^9 | A^{11} $G^{0+7=7}$

G^7 | A^{11}

A* Search become optimal if it follow the properties of h- admissibility

h - admissibility means

if heuristic cost can be underestimated or less by / from actual cost then that graph is admissible.

Actual cost = F to I path cost = 211

Heuristic cost of F = 178

∴ 178 < 211

So when for every node the heuristic value is less or is understimated by actual cost then A* search is optimal & admissible.

Fringe : (After change value from table {H-140})

A^{366} to = 366

A^{366}

B^{449}	C^{447}	E^{393}

E^{393}

B^{449}	C^{447}	G^{413}	F^{417}

G^{413}

B^{449}	C^{447}	F^{417}	H^{457}

F^{417}

B^{449}	C^{447}	H^{457}	I^{450}

C^{447}

B^{449}	H^{457}	I^{450}	D^{473}

B^{449}

H^{457}	I^{450}	D^{473}

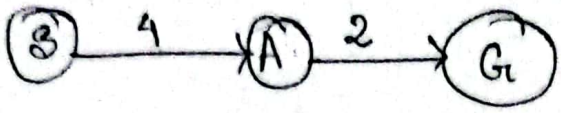
I^{450}

H^{457}	D^{473}

$A \rightarrow E \rightarrow F \rightarrow I$ (Not optimal)

As the heuristic here not admissible

H- Consistency :



3	5
A	2
0	

→ it can be $(n \leq 6)$

$$h(A) \leq c(A, G) + h(G)$$
$$2 + 0 = 2$$

$$h(S) \leq c(S, A) + h(A)$$
$$4 + 2 = 6$$

→ Consistent condition

that is the $h(n)$ of a particular node need to be less or equal from its path cost of succesor value and its succesor's heuristic value

ie $h(n) \leq c(n, n+1) + h(n+1)$

(S) → (A)

Now $h(S) = c(S, A) + h(A)$

$= 3 + 2$

$= 5$

A	2
S	6
G	0

The graph is admissible

$h(A) \leq c(A,G) + h(G)$

$7+0 = 7$

$h(S) \leq c(S,A) + h(A)$

$2+2 = 4$ (condition not fulfilled)

↳ so not consistant

If a graph is consistant it must be an it is always admissible.

If the graph is not consistant → It may have many irrelevant branching to reach the goal.

Heuristic Dominance:

$h_2 \geq h_1$

for each and every node h_2 returns large value than h_1 $\therefore h_2$ dominate h_1

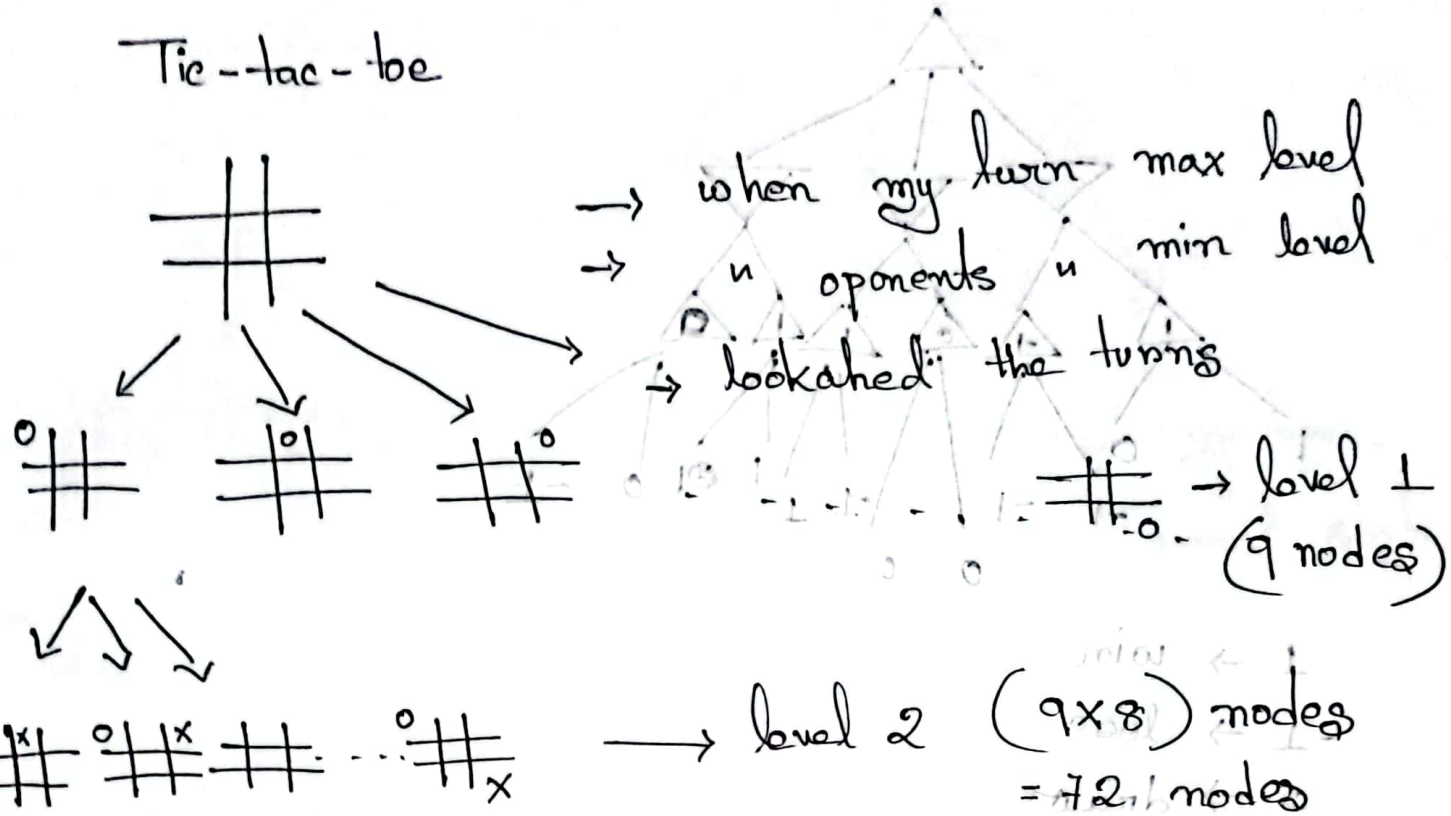
- large value should be picked

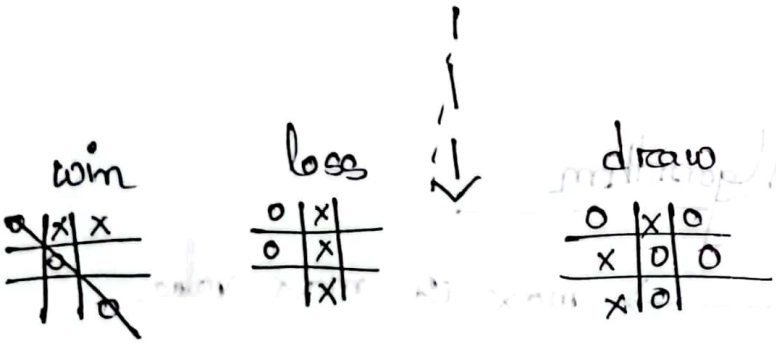
Quiz-1
Informed Search
(Saturday)

(Lecture-5)

Games → 2 player games

Tic--tac-toe





zero-sum game :

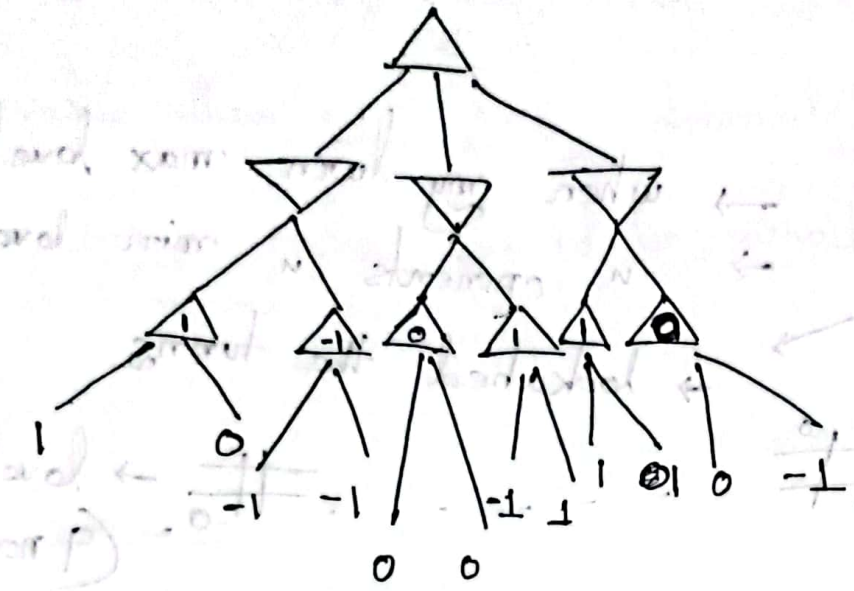
Adding gain and loss will be equal 0

If ones winning probability is maximized
other's " " " minimized

State Space (From my perspective):

max level → △

min level → ▽

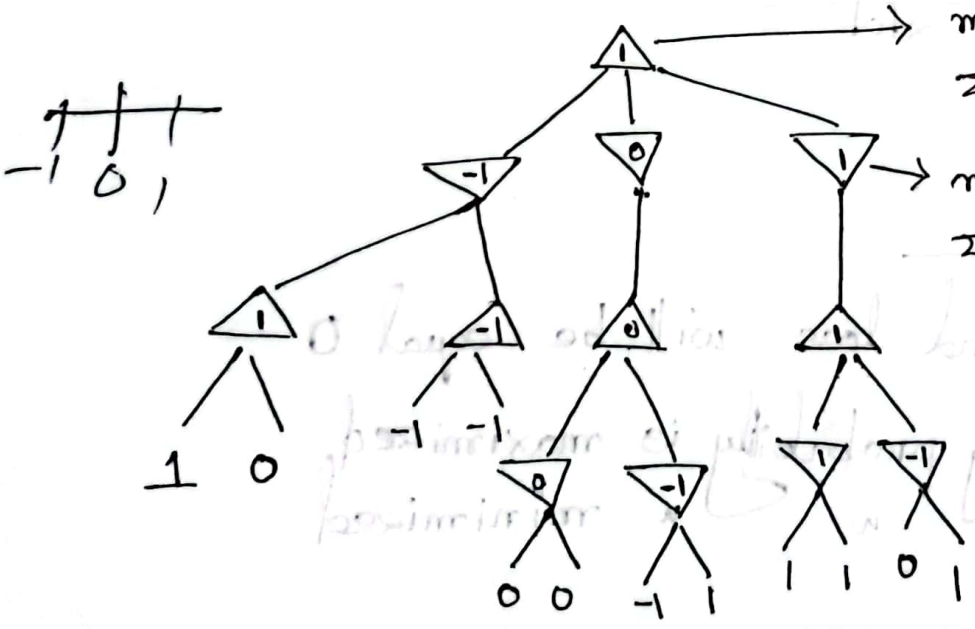


1 → win

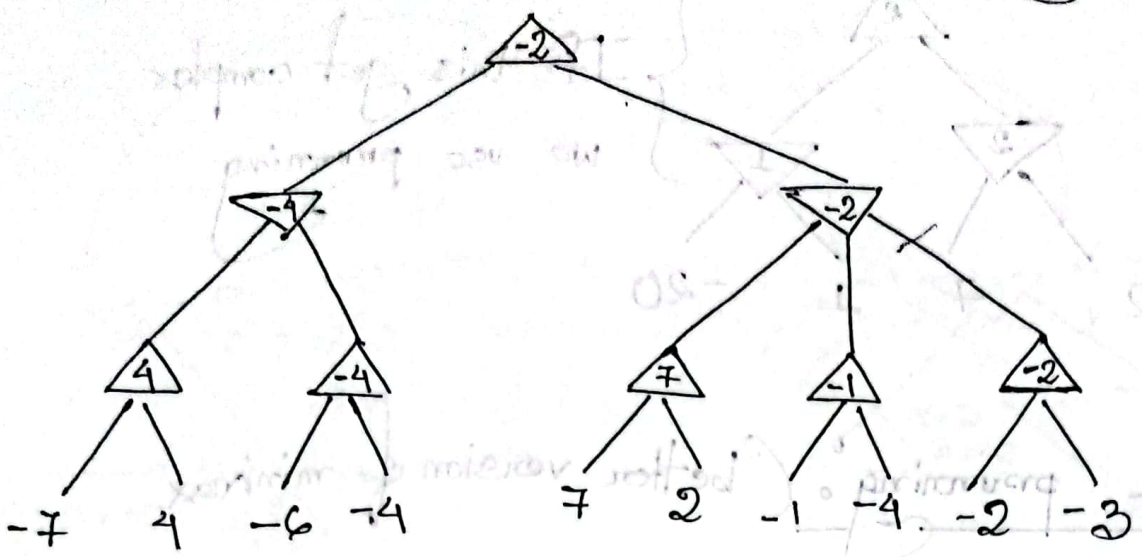
-1 → loss

0 → draw

Minimax Algorithm

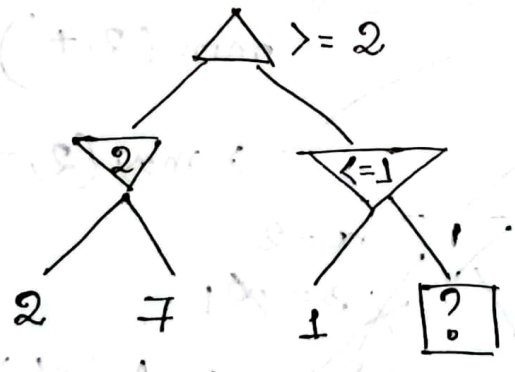


Returned value of a good function by minimax Algorithm

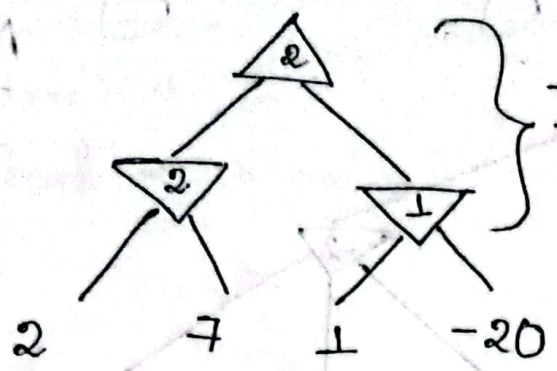


good function → mainly return postive / negetive response evalting the board state / current board state.

Intiution of value :



(Analysing the overall state it can be said that this value (?) will not effect the overall state)



If this get complex we use prunning

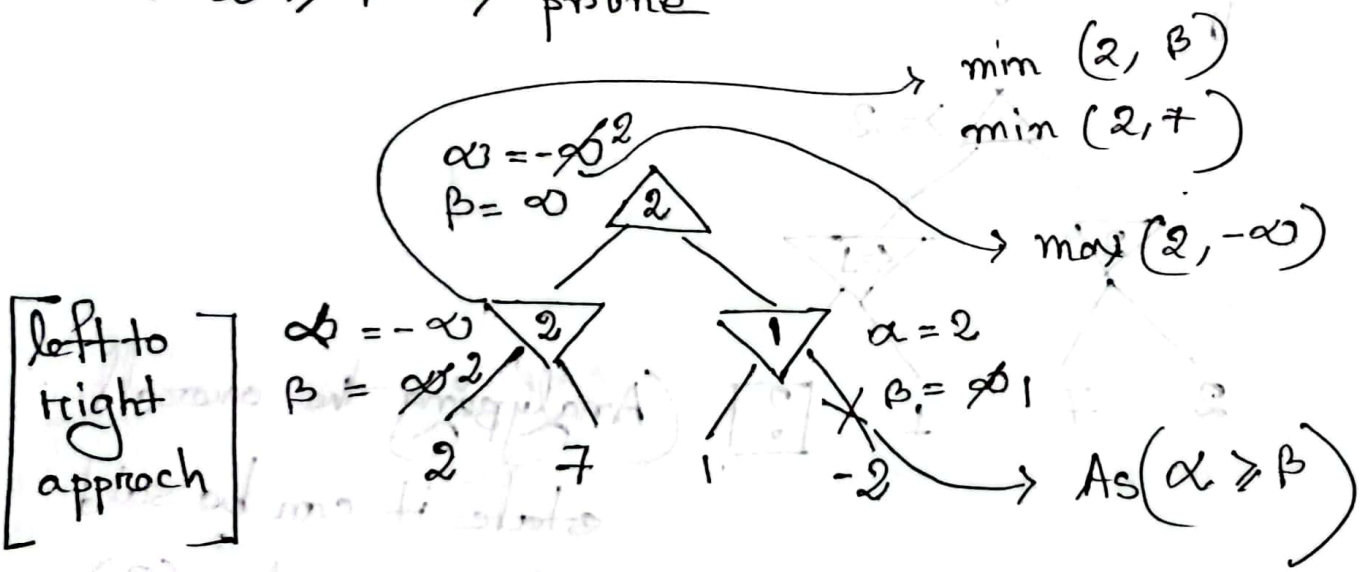
Alpha- beta prunning : (better version of minmax

$\alpha \rightarrow -\infty$
$\beta \rightarrow \infty$

- For max node → update α → max value
- " min " → " β → min value

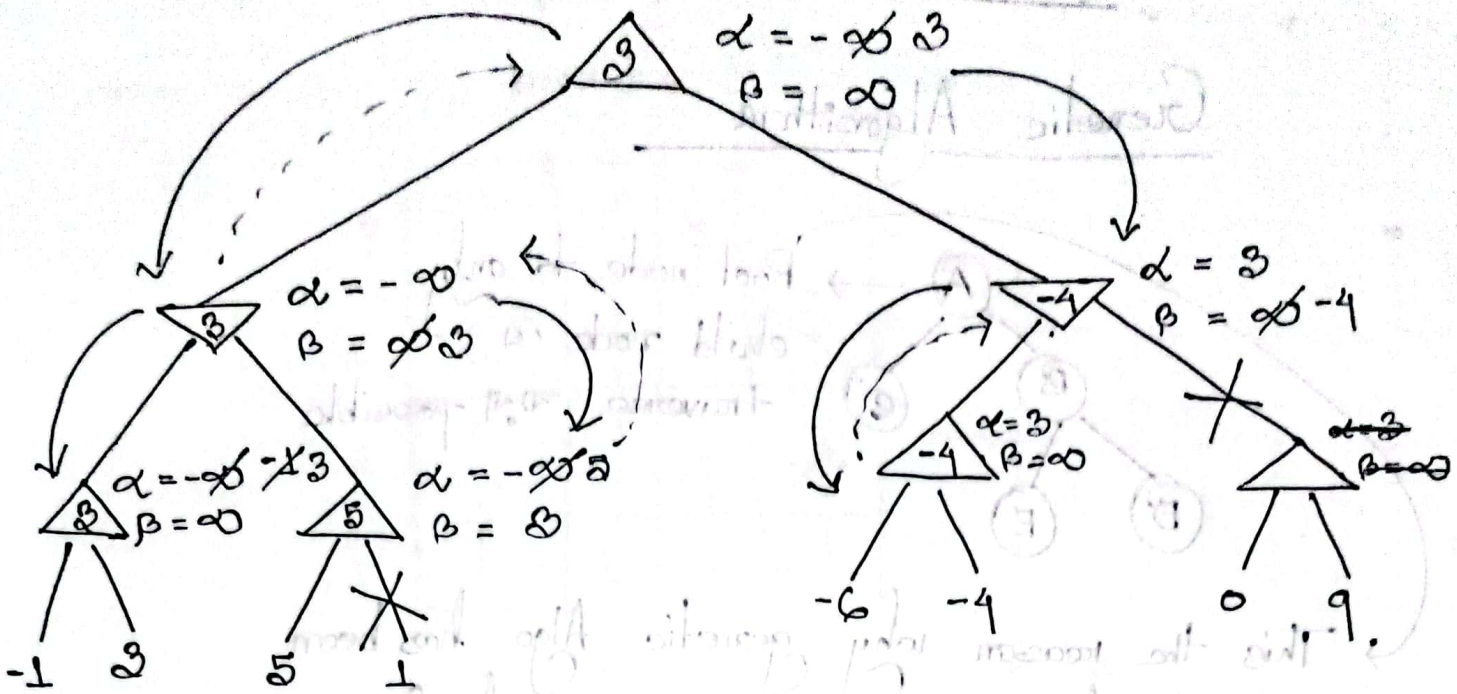
When,

- $\alpha \geq \beta$ → prune



- If the state tree is large the extra branch get cut out so become less expensive and reduce time

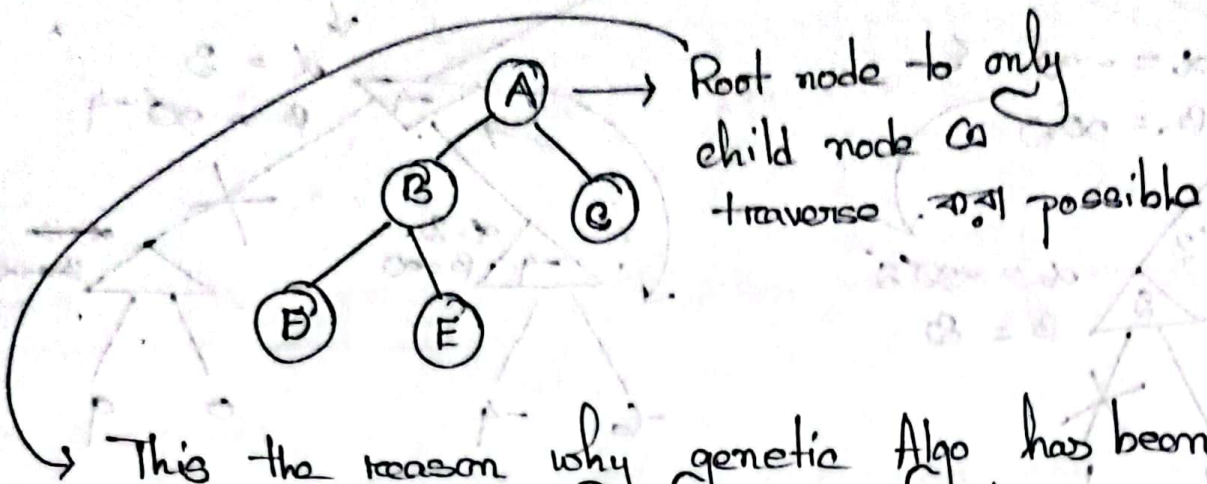
Example :



Visit → Alpha-beta pruning Stimulation (2nd page)

(Lecture-6)

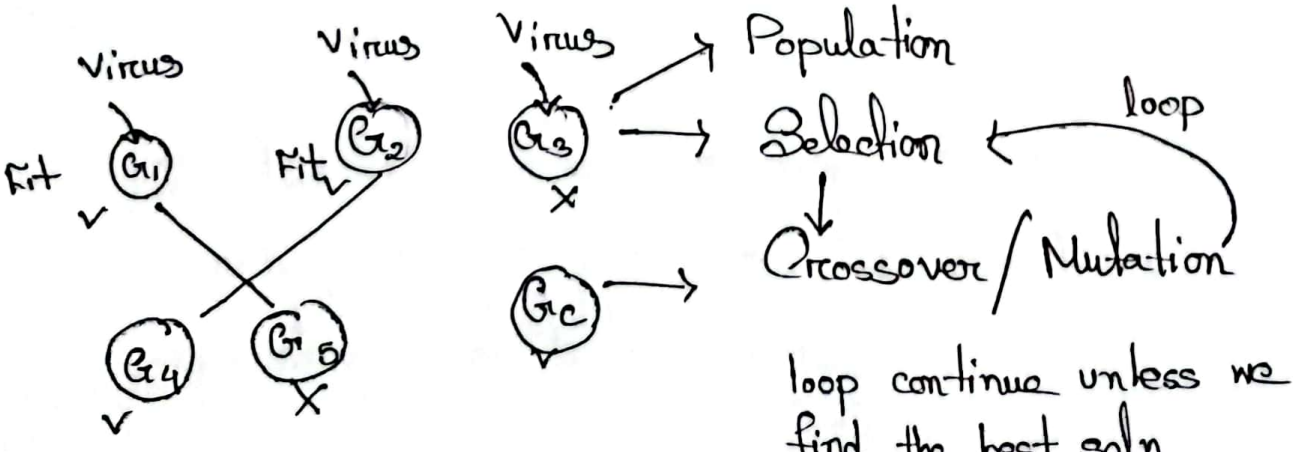
Genetic Algorithm



This the reason why genetic Algo has been introduced. As here the traversal from one state to another doesn't need to be its adjacent state / child node.

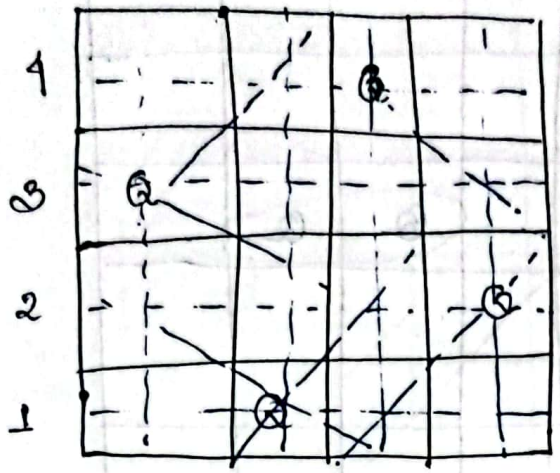
- It traverse, completely to different state from one state
- Work with only fit state.

Let create a vaccine for a disease using a immune gene.



n - queen problem

When $n = 4$, so board $= n^2 = 4^2 = 16$



3 1 4 2 — State

Let take $n = 8$

∴ For this we will take 4- state board so we will get 4 State :

Gene ← (2) 4 7 4 8 5 5 2 → chromosome 1 (Each element of chromosome/ sample is gene)

3 2 7 5 2 4 1 1 → 2

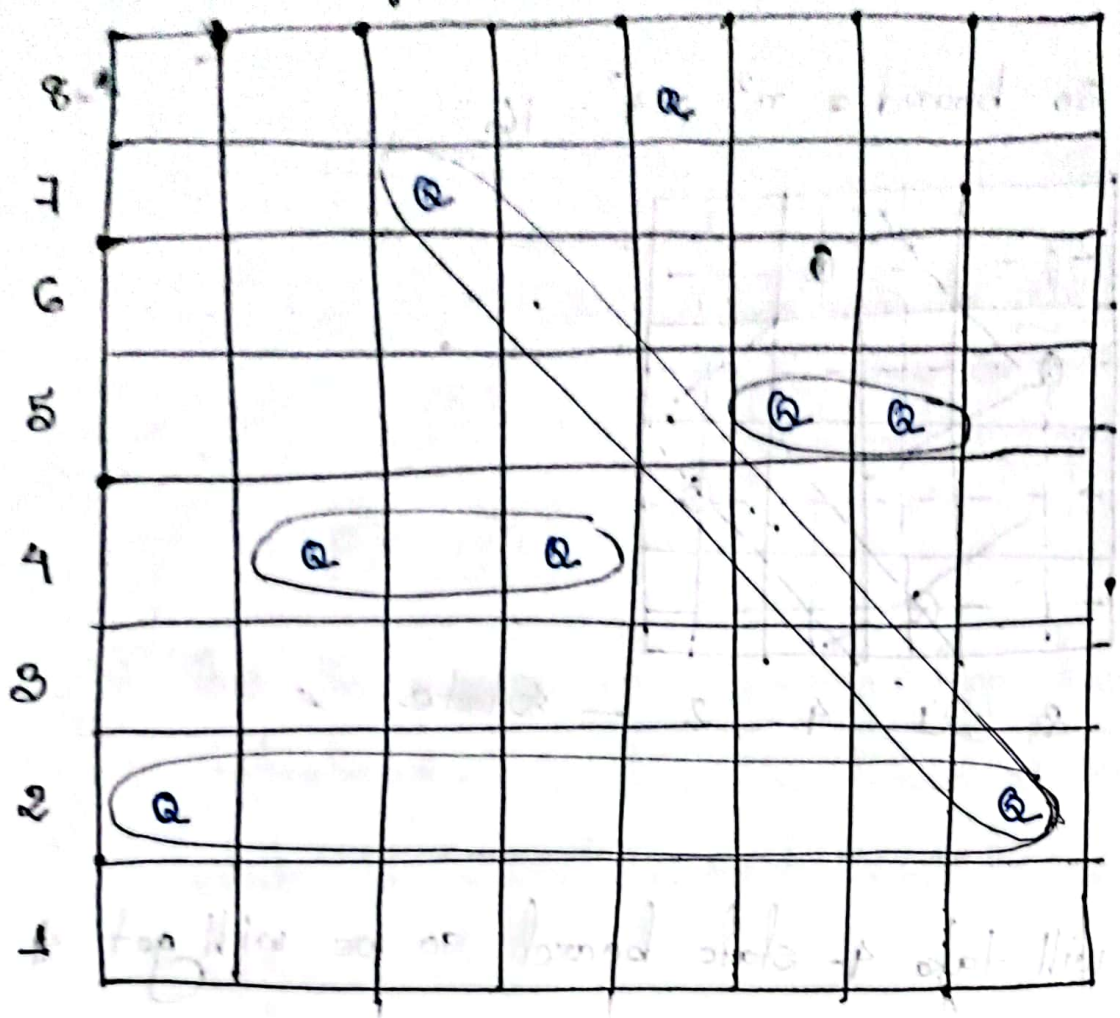
2 4 4 1 5 1 2 4 → 3

8 2 5 4 3 2 1 3 → 4

Fitness Function = no of non-clashing pair

↳ whose non clashing pair is more, that board is more fit

Board Stat of chromosome 1 :



So clashing pair is = 4

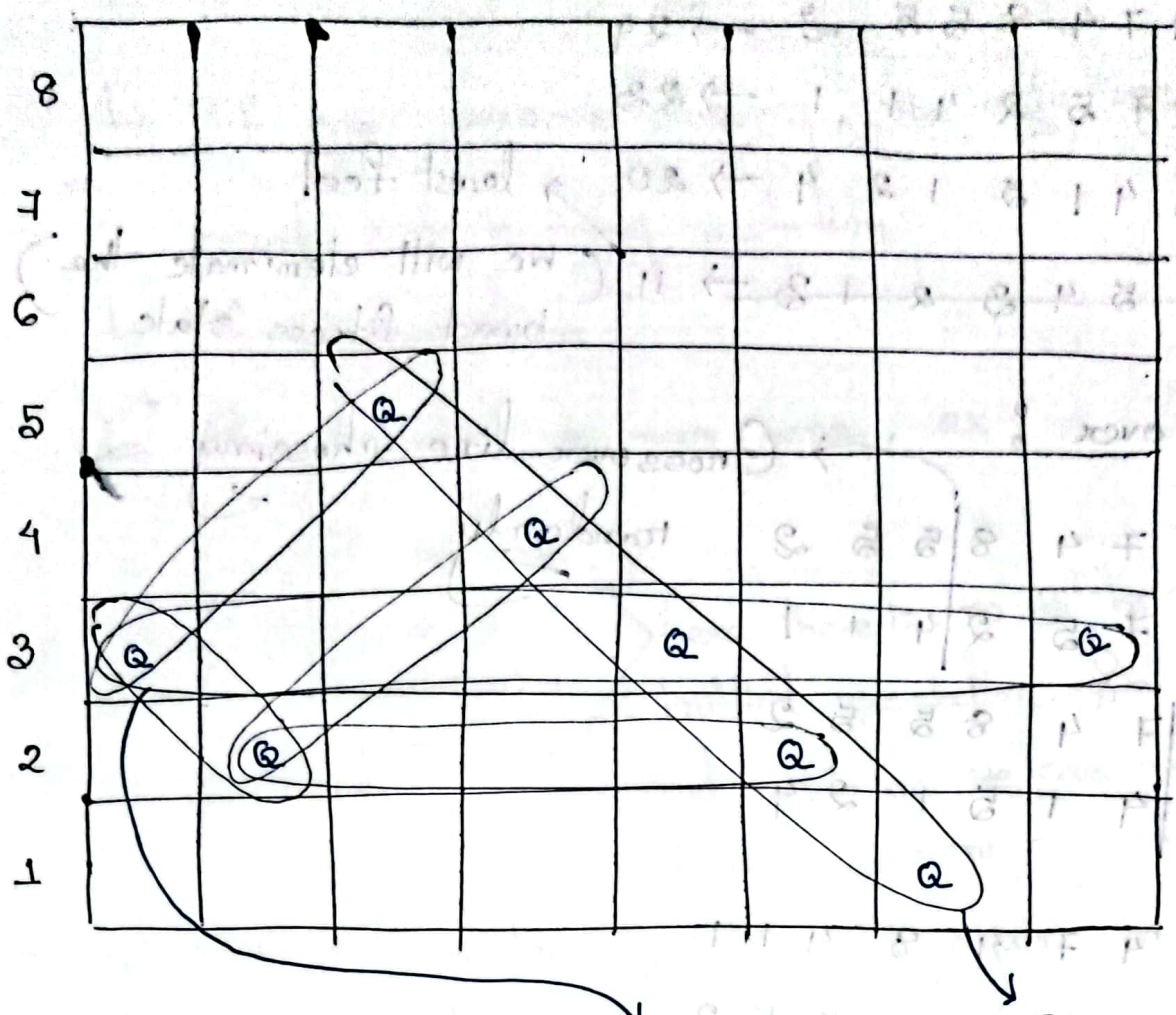
Total Pair = $^8C_2 = 28$

Non clashing pair = $28 - 4 = 24$

$^2C_2 + {}^2C_2 + {}^2C_2 + {}^2C_2$ → { $^2C_2$ because pairing between 2 }

$= 1 + 1 + 1 + 1$

Board State of Chromosome 1 :



No of clashing pair = $1 + {}^{3}C_2 + 1 + 1 + 1 + {}^{5}C_2$

$= 1 + 3 + 3 + 10 = 17$

∴ No of non clashing pair $= 28 - 17 = 11$

→ So for n-Queen making pair, total pair = $\boxed{{}^{n}C_2}$.

Step 1

So the fitness of the States are :

2 4 7 4 8 5 5 2 → 24

8 2 7 5 2 4 1 1 → 23

2 4 4 1 5 1 2 4 → 20 → least feet

~~3 2 5 4 3 2 1 3~~ → 11 (We will elemimate the lowest fitness state)

Step 2

Cross over :

→ Crossover line chossing randomly

2 4 7 4 8 | 5 5 2

8 2 7 5 2 | 4 1 1

2 4 | 7 4 8 5 5 2

2 4 | 4 1 5 1 2 4

2 4 7 4 8 4 1 1

8 2 7 5 2 5 5 2

2 4 4 1 5 1 2 4

2 4 7 4 8 5 5 2

Step 3

Mutation :

2 4 7 4 (8) 4 1 1 → 2 4 7 4 5 4 1 1

8 (2) 7 5 2 5 5 2 → 8 8 7 5 2 5 5 2

2 4 4 1 5 1 2 4 → 2 4 4 1 5 1 2 4

2 4 7 4 8 5 5 (2) → 2 4 7 4 8 5 5 2

Why Mutation ?

↳ To diversified the state

↳ If any gene is missing, the only way to insert that is doing mutation

↳ To remove clash

↳ Can introduce a new gene, ex-

1	4	4	2
1	8	2	1
2	4	2	4
4	1	1	4

here there is no 3 in the initial population. So without mutation we can never induce 3 here.

This 3 step will reactead again and again in a loop, unless we get the desire result.

(Lecture- 7)

0/1 knapsack Problem

↳ হয় নিবে else বাদ দিবে।

Steps:
- Selection
- Crossover
- Mutation

Selection of Chromose for knapsack:

A B C D
C D
A B
A E F Gr H

As there is chromosome of different length, the crossover operation will problematic.

↳ So to solve the problem, we will encode chomosome with binary number.

	A	B	C	D	E	F	Gr	H
C_1	1	1	1	1	0	0	0	0
C_2	0	0	1	1	0	0	0	0
C_3	1	1	0	0	0	0	0	0
C_4	1	0	0	0	1	1	1	1

fitness function will not work here as n-Queen

∴ Here fitness = Total reward

if $C_1(w) >$ max-weight

$f(c_1) = 0$

else

$f(c_1) = C_1(R)$

Fitness :

$c_1 = f(c_1) = 75 > 12 \therefore f(c_1) = 0$ As $C_1(w) > 12$

$c_2 = f(c_2) = 50 > 12 \therefore f(c_2) = 0$ " $C_2(w) > 12$

$c_3 = f(c_3) = 25 > 12 \therefore f(c_3) = 25$ " $C_3(w) < 12$

$c_4 = f(c_4) = 25 > 12 \therefore f(c_4) = 0$ " $C_4(w) > 12$

incase all become 0 we will take the lowest one.

As tie occure will use different approach.

$f(c_1) = 0 - |\text{max-weight} - C_1(w)|$

$= |12 - 14| = 2$

$f(c_2) = |12 - 19| = 7$

$f(c_3) = |12 - 3| = 25$

$f(c_4) = |12 - 19| = -7$

After that do crossover & mutation

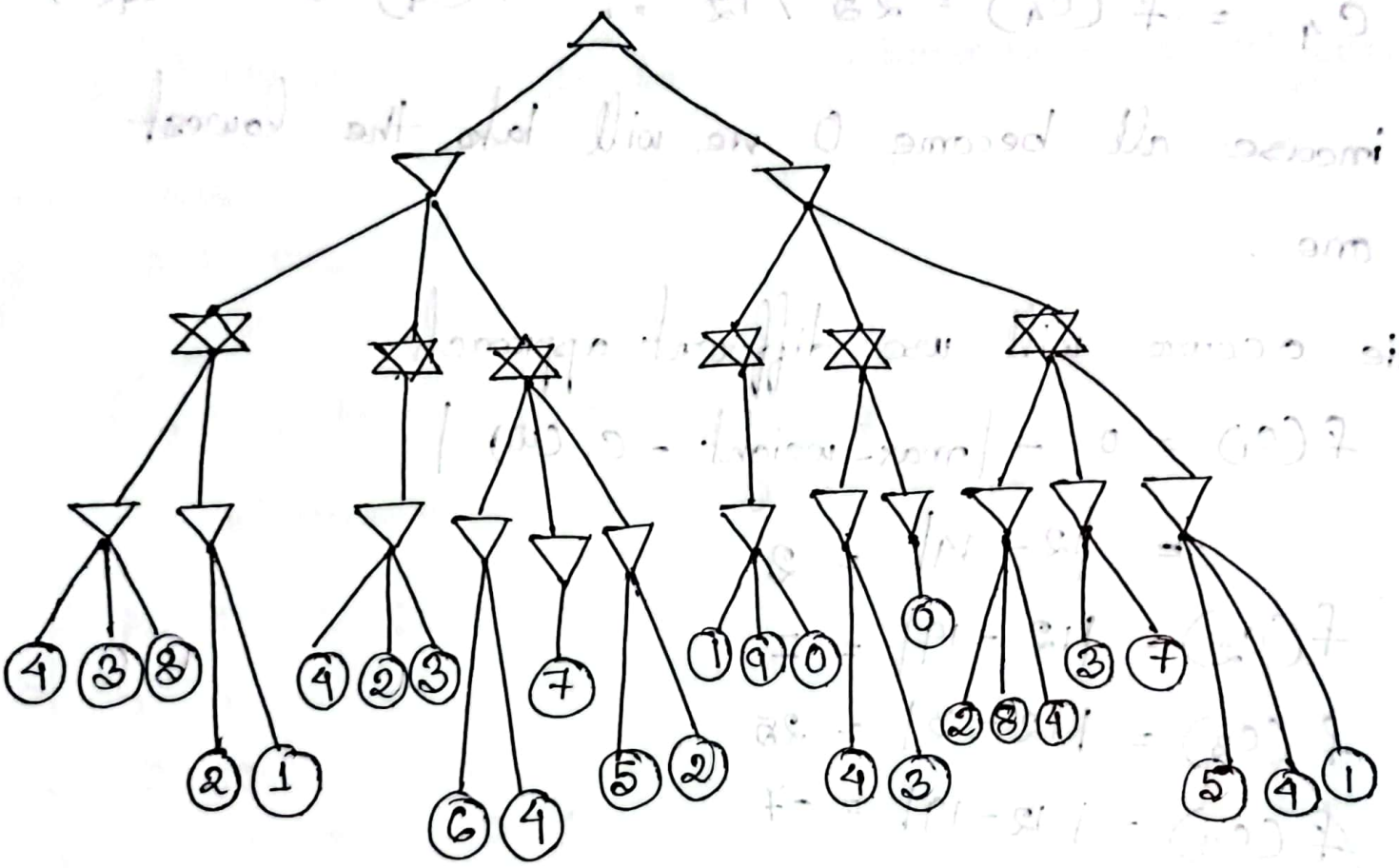
Problem :

$$(x_2 + x_4) - (x_1 + x_3) = 50$$

x_1, x_2, x_3, x_4

$$\therefore F(c_1) = \left|(x_2 + x_4) - (x_1 + x_3)\right| - 50$$

Alfa - Beta Pruning



Quiz 2 → Games

Quiz 3 → Genetic Algorithm (Next Thursday)